

One of my patient, who was admitted 8 days back and currently being discharged today

Initial Assessment was entered by night duty resident

Step 1—copied directly from EMR system of Hospital to ChatGPT---This gives AI the Clinical history of Patient

Diagnosis

- Hemoptysis – evaluation for NOAC-induced bleeding vs infective etiology vs autoimmune etiology
 - Suspected diffuse alveolar hemorrhage (DAH)
 - Acute kidney injury on chronic kidney disease (baseline serum creatinine ~1.5 mg/dL)
 - Dementia, likely vascular
 - Coronary artery disease (2016)
 - Cerebrovascular accident (2019)
 - Intermittent atrial fibrillation on anticoagulation
 - Hypertension
-

Presenting Complaints

- Breathlessness for 3–4 days
 - Hemoptysis for 2 days
 - Fever since 1 day
 - Excessive sputum production for 2–3 months
 - Forgetfulness and irrelevant talks for past few months
-

History of Present Illness

Mr. Rajinder Nath Sharma, an 81-year-old male, presented with complaints of breathlessness for the past 3–4 days associated with hemoptysis for 2 days and fever since the previous day. He also had a history of excessive sputum production for the preceding 2–3 months.

The patient had been experiencing progressive forgetfulness and episodes of irrelevant speech over the past few months and was undergoing evaluation for cognitive decline. Previous evaluation had reportedly revealed greater than 80% internal carotid artery stenosis, raising the possibility of vascular dementia.

Prior to MICU admission, the patient was being evaluated by the Cardiology team. HRCT chest showed bilateral basal ground-glass opacities with subpleural sparing, raising suspicion

of diffuse alveolar hemorrhage. In view of ongoing hemoptysis and abnormal imaging findings, the patient was shifted to the Medical ICU for further evaluation and management.

Past History

- Coronary artery disease (2016)
 - Cerebrovascular accident (2019)
 - Intermittent atrial fibrillation
 - Chronic kidney disease with baseline serum creatinine approximately 1.5 mg/dL
 - Suspected vascular dementia under evaluation
 - Hypertension
-

Medication History

- Tab Eliquis (Apixaban)
 - Tab Ecosprin (Aspirin)
 - Tab Avas (Atorvastatin)
 - Tab Betaloc (Metoprolol)
 - Tab Telma (Telmisartan)
-

Presenting Status

- Conscious and alert
 - Mild tachypnea present
 - Hemodynamically stable
 - Maintaining SpO₂ 97% on room air
-

Plan

- Autoimmune workup including ANA, ANCA and p-ANCA
- Urine routine examination and culture
- Blood sugar monitoring (pre-meal RBS)
- Started on Inj Tazact
- Antiplatelet and anticoagulant medications kept on hold in view of hemoptysis
- Further evaluation and management in MICU

Patient was being evaluated and managed in ICU and his daily progress notes were provided to ChatGPT as voice notes

Step-2—Providing Daily progress notes and Vital Investigations to ChatGPT as follows

progress notes over days.....Patient Rajendra Nath Sharma, he is a known case of CAD post-CABG in 2016, history of CVA in 2019. Patient was on oral anticoagulation. He also had CKD. His baseline creatinine in 2025 was 1.5, which gradually increased to 1.8 in December 2025, and presently on admission he had creatinine of 2.5. He, after initial presentation, he was transferred under pulmonology team because of CT chest showing bilateral perihilar and basal areas GGOs, suggestive of possible diffuse alveolar hemorrhage as on admission he was having significant hemoptysis. His, in view of deranged RFT and hemoptysis, possibility of underlying autoimmune process was considered and ANA by immunofluorescence was sent, which was reported significantly positive. In view of positive ANA, he was further evaluated and ENA panel was sent, which was reported negative. Also, sputum sample was sent for multiplex PCR and Gram staining and culture. It also came out negative. Also, possibility of renal pulmonary disease was considered and hence anti-GBM antibody was also sent. It also was reported negative.

These were Silently Integrated by ChatGPT into Patients profile

Further Voice notes as follows

Nephrology opinion was also taken for Rajendranath Sharma and urine ACR was done, which showed sub-nephrotic range proteinuria. There was no RBCs in urine. So possibility of CKD was predominantly considered in this patient and renal biopsy was not warranted. However, for management of diffuse alveolar hemorrhage, his oral anticoagulation was continued to be stopped and he was continued on injection solumedrol 40 milligram twice daily. Gradually his clinical condition improved, his hemoptysis settled. A repeat HRCT chest was done, which showed slight decrease in alveolar opacities. So after stabilization, he has been shifted to ward on 3rd June 2026.

Further Voice Notes

Now he has stabilised around day 7, there is no hemoptysis and radiology has also improved and is planned for discharge with advice to follow up in OPD after 7 days

Step 3:-Now I want Final Detailed Course in Hospital so that I can Copy and enter into Discharge Format

What the Final AI-Generated "Course in Hospital" Would Look Like

This is the key example for engineers because the final narrative is generated from multiple small updates entered over 7 days:

Course in Hospital

The patient, a known case of coronary artery disease status post CABG (2016), cerebrovascular accident (2019), chronic kidney disease and intermittent atrial fibrillation on oral anticoagulation, was admitted with complaints of breathlessness, hemoptysis, fever and excessive sputum production. HRCT chest demonstrated bilateral basal and perihilar ground-glass opacities with subpleural sparing, raising suspicion of diffuse alveolar hemorrhage. In view of ongoing hemoptysis and abnormal imaging findings, the patient was transferred to the Pulmonology service for further evaluation and management.

Considering the coexistence of hemoptysis and worsening renal dysfunction, a pulmonary-renal syndrome was suspected. Autoimmune workup revealed significantly positive ANA by immunofluorescence, while ENA panel and anti-GBM antibodies were negative. Infectious workup including sputum multiplex PCR, Gram stain and culture was negative.

Nephrology opinion was obtained. Urine ACR showed sub-nephrotic range proteinuria without hematuria or active urinary sediment. Renal dysfunction was considered predominantly secondary to underlying chronic kidney disease, and renal biopsy was not considered indicated.

Oral anticoagulation was withheld in view of significant hemoptysis, and the patient was managed with intravenous methylprednisolone. The patient showed gradual clinical improvement with progressive resolution of hemoptysis. Repeat HRCT chest demonstrated a decrease in bilateral alveolar opacities, indicating radiological improvement.

Following clinical stabilization and improvement in respiratory status, the patient was shifted from MICU to the ward on 03 June 2026. During the subsequent hospital stay, he remained stable without recurrence of hemoptysis. In view of sustained clinical and radiological improvement, the patient was discharged in stable condition with advice for follow-up in the Pulmonology Outpatient Department after 7 days.

Step-4 Actual Discharge Summary Looks Like Following (Copied from Actual Summary Given to patient)

DISCHARGE SUMMARY

Department of Pulmonary Medicine

Name of Patient:-

Age/Sex

Max ID:-

Address

Date of admission

Date of Discharge

Primary Consultant:-

Diagnosis

Primary Diagnosis

- Suspected diffuse alveolar hemorrhage presenting with hemoptysis, clinically and radiologically improved following corticosteroid therapy

Secondary Diagnoses

- Chronic kidney disease with acute worsening of renal function
- Coronary artery disease, status post CABG (2016)
- Intermittent atrial fibrillation
- Hypertension
- Cerebrovascular accident (2019)
- Suspected vascular dementia under evaluation

Conditions Evaluated During Hospitalization

- Pulmonary-renal syndrome – investigated, not confirmed
- Anti-GBM disease – ruled out
- Infectious etiology – no microbiological evidence identified

Presenting Complaints

- Breathlessness for 3–4 days
- Hemoptysis for 2 days

- Fever for 1 day
 - Excessive sputum production for 2–3 months
 - Progressive forgetfulness and episodes of irrelevant speech over the past few months
-

History of Present Illness

Mr. Rajinder Nath Sharma, an 81-year-old gentleman, known case of coronary artery disease status post CABG (2016), cerebrovascular accident (2019), intermittent atrial fibrillation on oral anticoagulation, chronic kidney disease, and hypertension, presented with complaints of breathlessness for 3–4 days associated with hemoptysis for 2 days and fever for 1 day. He also reported excessive sputum production for the preceding 2–3 months.

The patient had been experiencing progressive forgetfulness and episodes of irrelevant speech over the preceding months and was undergoing evaluation for possible vascular dementia. Previous evaluation reportedly demonstrated significant carotid artery stenosis.

HRCT chest performed during evaluation demonstrated bilateral perihilar and basal ground-glass opacities suggestive of diffuse alveolar hemorrhage, following which he was admitted for further evaluation and management.

Past History

- Coronary artery disease, status post CABG (2016)
 - Cerebrovascular accident (2019)
 - Intermittent atrial fibrillation
 - Chronic kidney disease
 - Hypertension
 - Suspected vascular dementia under evaluation
-

Medication History

- Tab Eliquis (Apixaban)
 - Tab Ecosprin (Aspirin)
 - Tab Avas (Atorvastatin)
 - Tab Betaloc (Metoprolol)
 - Tab Telma (Telmisartan)
-

Presenting Status

At admission, the patient was conscious and alert. Mild tachypnea was present. Oxygen saturation was maintained at 97% on room air. He was hemodynamically stable but had ongoing significant hemoptysis.

Significant Investigations

HRCT Chest

- Bilateral perihilar and basal ground-glass opacities with findings suggestive of diffuse alveolar hemorrhage.

Repeat HRCT Chest

- Interval decrease in alveolar opacities suggestive of radiological improvement.

Autoimmune Workup

- ANA by Immunofluorescence: Significantly positive.
- ENA Panel: Negative.
- Anti-GBM Antibody: Negative.

Renal Evaluation

- Baseline serum creatinine (2025): ~1.5 mg/dL.
- December 2025: ~1.8 mg/dL.
- Admission serum creatinine: 2.5 mg/dL.
- Urine ACR: Sub-nephrotic range proteinuria.
- No hematuria or active urinary sediment.

Infectious Workup

- Sputum Multiplex PCR: Negative.
 - Sputum Gram Stain and Culture: Negative.
-

Course in Hospital

The patient was admitted with hemoptysis, breathlessness, and bilateral pulmonary infiltrates. HRCT chest demonstrated bilateral perihilar and basal ground-glass opacities suggestive of diffuse alveolar hemorrhage. In view of significant hemoptysis and radiological findings, he was transferred under the Pulmonology service for further evaluation and management.

The patient was a known case of chronic kidney disease with progressively worsening renal function. Considering the coexistence of hemoptysis and renal dysfunction, the possibility of

an underlying pulmonary-renal syndrome was considered and a comprehensive autoimmune evaluation was initiated.

ANA by immunofluorescence was reported significantly positive. Further evaluation with ENA panel and Anti-GBM antibodies was negative. Simultaneously, infectious causes were evaluated through sputum multiplex PCR, Gram stain, and culture, all of which were negative.

Nephrology opinion was sought. Urine ACR revealed sub-nephrotic range proteinuria without hematuria or active urinary sediment. In view of the overall clinical picture, renal dysfunction was considered predominantly attributable to underlying chronic kidney disease rather than active glomerulonephritis. Renal biopsy was not considered indicated.

Oral anticoagulation was withheld because of ongoing hemoptysis. The patient was treated with intravenous methylprednisolone (Solu-Medrol) 40 mg twice daily. Over the subsequent days, he demonstrated progressive clinical improvement with complete resolution of hemoptysis.

Repeat HRCT chest demonstrated interval reduction in alveolar opacities, indicating radiological improvement. Following stabilization, the patient was transferred from the Medical ICU to the ward on 03 June 2026.

During the subsequent ward stay, he remained clinically stable without recurrence of hemoptysis. Respiratory symptoms improved significantly and chest imaging demonstrated substantial clearing of infiltrates. In view of sustained clinical and radiological improvement, he was considered fit for discharge.

Cross-Consultations

Nephrology

Consultant Name:-

- Evaluated for pulmonary-renal syndrome.
- Urine ACR showed sub-nephrotic range proteinuria.
- No hematuria or active urinary sediment.
- CKD considered the predominant cause of renal dysfunction.
- Renal biopsy not indicated.

Medication Changes During Admission

Medications Held

- Eliquis (Apixaban) withheld because of significant hemoptysis and suspected diffuse alveolar hemorrhage.

New Medications Initiated

- Intravenous methylprednisolone therapy for suspected diffuse alveolar hemorrhage.

Important Note

- Anticoagulation status to be reassessed during follow-up based on clinical recovery and bleeding risk assessment.
-

Current Status at Discharge

- Conscious and oriented.
 - Hemodynamically stable.
 - Maintaining oxygen saturation of 97% on room air.
 - No tachypnea.
 - No hemoptysis.
 - Chest X-ray demonstrates significant radiological improvement.
 - Ambulating independently.
 - Clinically stable for discharge.
-

Functional Status at Discharge

- Ambulating independently.
 - Maintaining oxygenation on room air.
 - Oral intake adequate.
 - No requirement for supplemental oxygen.
 - Independent in routine daily activities.
-

Discharge Medications

1. **Tab Ceftum 500 mg** – One tablet twice daily.
 2. **Tab Pan 40 mg** – One tablet once daily before breakfast.
 3. **Tab Ecosprin AV 150/20 mg** – One tablet once daily.
 4. **Tab CCM** – One tablet once daily.
 5. **Nebulization with Dolin** – Three times daily.
 6. **Tab Risalone 40 mg** – One tablet once daily until reviewed in OPD.
-

Follow-Up Plan

- Follow up in Pulmonology OPD after 7 days.
 - Review clinical status and recurrence of hemoptysis, if any.
 - Reassess corticosteroid requirement and tapering schedule.
 - Reassess anticoagulation strategy in view of atrial fibrillation and prior hemoptysis.
 - Repeat CBC, renal function tests, and chest imaging as clinically indicated.
-

Pending Results

- No pending investigation results at the time of discharge.
-

When to Seek Emergency Medical Help

The patient and family have been advised to seek immediate medical attention or report to the Emergency Department in case of:

- Blood in sputum (hemoptysis)
- Increase in cough
- Worsening breathing difficulty
- Chest pain
- Any sudden deterioration in clinical condition

Emergency Contact Number: 0172-5212-000

Discharge Condition

Patient discharged in stable condition with complete resolution of hemoptysis, significant radiological improvement, and advice for close outpatient follow-up.